pMapper: Power and Migration Cost Aware Application Placement in Virtualized Systems

Akshat Verma[1], Puneet Ahuja[2], and Anindya Neogi[1]

[1] IBM India Research Lab
[2] IIT Delhi

Abstract. Workload placement on servers has been traditionally driven by mainly performance objectives. In this work, we investigate the design, implementation, and evaluation of a power-aware application placement controller in the context of an environment with heterogeneous virtualized server clusters. The placement component of the application management middleware takes into account the power and migration costs in addition to the performance benefit while placing the application containers on the physical servers. The contribution of this work is two-fold: first, we present multiple ways to capture the cost-aware application placement problem that may be applied to various settings. For each formulation, we provide details on the kind of information required to solve the problems, the model assumptions, and the practicality of the assumptions on real servers. In the second part of our study, we present the pMapper architecture and placement algorithms to solve one practical formulation of the problem: minimizing power subject to a fixed performance requirement. We present comprehensive theoretical and experimental evidence to establish the efficacy of pMapper.

1 Introduction

Resource provisioning or placement of applications on a set of physical servers to optimize the application Service Level Agreements (SLA) is a well studied problem [6,24]. Typically, concerns about application performance, infrequent but inevitable workload peaks, and security requirements persuade the provisioning decision logic to opt for a conservative approach, such as hardware isolation among applications with minimum sharing of resources. This leads to sub-optimal resource utilization. Bohrer et al. have studied real webserver workloads from sports, e-commerce, financial, and internet proxy clusters to find that average server utilization varies between 11% and 50% [3]. Such inefficient provisioning leads to relatively large hardware and operations costs when compared to the actual workload handled by the data center. However, recently two important trends, *viz.* server virtualization and the heightened awareness around energy management technologies, have renewed interest in the problem of application placement. The placement logic in the middleware now need to look beyond just application SLAs into increasing energy-related operations costs. In this paper, we investigate the *Power-aware Application Placement* problem and

V. Issarny and R. Schantz (Eds.): Middleware 2008, LNCS 5346, pp. 243–264, 2008.